In [1]:
```
import sys
print(sys.executable)
#%cd C:\School\yolo5\yolov5
import os
print(os.getcwd())
#hi, please change directory to your yolov5 folder with the data files
import torch
print(torch.__version__)
print(torch.cuda.is_available())
print(torch.cuda.device_count())
print(torch.cuda.get_device_name(0) if torch.cuda.is_available() else "No
```

```
C:\Users\nothi\anaconda3\envs\yolov5\python.exe
C:\Users\nothi
2.7.1+cu118
True
1
NVIDIA GeForce RTX 4060 Laptop GPU
```

In [2]:
```
from ultralytics import YOLO
print("ultralytics imported ok")
```

```
ultralytics imported ok
```

In [9]:
```
model = YOLO('06-object-detection-using-yolo/yolov8n.pt')
```

```
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/
yolov8n.pt to '06-object-detection-using-yolo\yolov8n.pt': 100% ━━━━━━━━━━━━
5.1MB/s 1.2s.2s<0.0s.4s
```

In [10]:
```
#loads pretrained nano small model
model.info()
```

```
YOLOv8n summary: 129 layers, 3,157,200 parameters, 0 gradients, 8.9 GFLOPs
```

Out[10]:
```
(129, 3157200, 0, 8.8575488)
```

In [11]:
```
#should work if your notebook is running from the yolov5 folder.
results = model("data/images/bus.jpg")
```

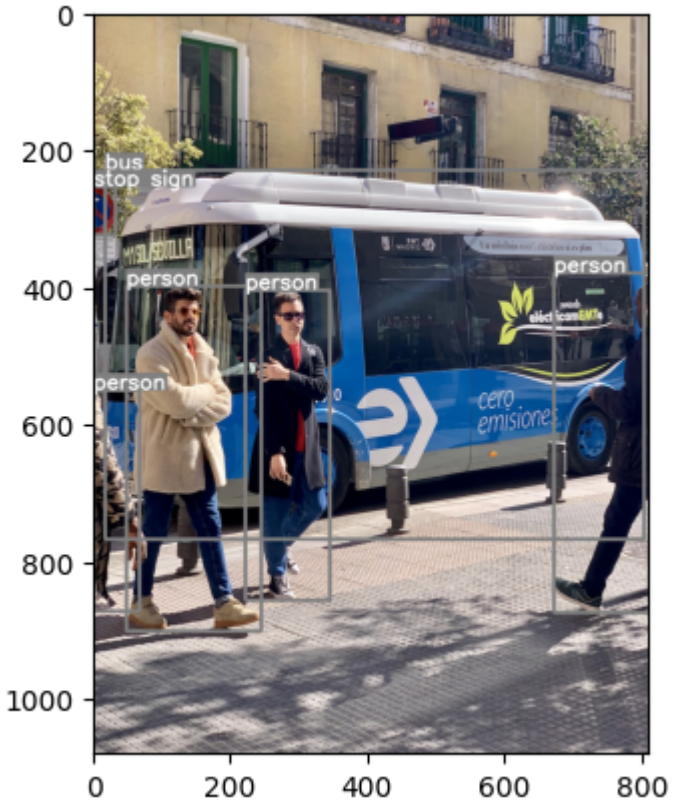
```
image 1/1 C:\School\yolo5\yolov5\data\images\bus.jpg: 640x480 4 persons, 1 bu
stop sign, 188.5ms
Speed: 11.8ms preprocess, 188.5ms inference, 24.6ms postprocess per image at
(1, 3, 640, 480)
```

In [12]:
```
from ultralytics.utils.plotting import Annotator
import matplotlib.pyplot as plt
import cv2

annotator = Annotator(cv2.cvtColor(results[0].orig_img, cv2.COLOR_BGR2RGB
boxes = results[0].boxes
for box in boxes:
    b = box.xyxy[0]  # get box coordinates in (left, top, right, bottom)
    c = box.cls
    annotator.box_label(b, model.names[int(c)])

plt.imshow(annotator.result())
```

Out[12]:
```
<matplotlib.image.AxesImage at 0x21e880bb2b0>
```



In [13]:
```
#this should work if you are in the folder, with relative path
from ultralytics import YOLO

model = YOLO("yolov8n.pt")
results = model("data/images/zidane.jpg")
```

```
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/
yolov8n.pt to 'yolov8n.pt': 100% ━━━━━━━━━━━━ 6.2MB 8.7MB/s 0.7s0.6s<0.1s1s

image 1/1 C:\School\yolo5\yolov5\data\images\zidane.jpg: 384x640 2 persons, 1
135.8ms
Speed: 4.4ms preprocess, 135.8ms inference, 1.9ms postprocess per image at sh
(1, 3, 384, 640)
```

In [14]:
```
from ultralytics.utils.plotting import Annotator
import matplotlib.pyplot as plt
import cv2

annotator = Annotator(cv2.cvtColor(results[0].orig_img, cv2.COLOR_BGR2RGB
boxes = results[0].boxes

for box in boxes:
    b = box.xyxy[0]
    c = box.cls
    annotator.box_label(b, model.names[int(c)])

plt.figure(figsize=(10, 8))
plt.imshow(annotator.result())
plt.axis("off")
plt.show()
```

In [16]:
```
#sees 2 people and a tie
#what not to do
#don't train from stratch
"""
from ultralytics import YOLO

model = YOLO('06-object-detection-using-yolo/yolov8n.yaml')  # Using yolo
model.info()
results = model(source='06-object-detection-using-yolo/zidane.jpg')

from ultralytics.utils.plotting import Annotator
import matplotlib.pyplot as plt
import cv2

annotator = Annotator(cv2.cvtColor(results[0].orig_img, cv2.COLOR_BGR2RGB
boxes = results[0].boxes
for box in boxes:
    b = box.xyxy[0]  # get box coordinates in (left, top, right, bottom)
    c = box.cls
    annotator.box_label(b, model.names[int(c)])

plt.imshow(annotator.result())
"""
from ultralytics import YOLO

model = YOLO('yolov8n.yaml')  # Using yolov8n structure, but randomly ini
model.info()
```

```
YOLOv8n summary: 130 layers, 3,157,200 parameters, 3,157,184 gradients, 8.9 G
```

Out[16]:
```
(130, 3157200, 3157184, 8.8575488)
```

In [21]:
```
#%cd C:\School\yolo5\yolov5
#do your relative path
results = model(source="data/images/zidane.jpg")

from ultralytics.utils.plotting import Annotator
```

```
import matplotlib.pyplot as plt
import cv2

annotator = Annotator(cv2.cvtColor(results[0].orig_img, cv2.COLOR_BGR2RGB
boxes = results[0].boxes
for box in boxes:
    b = box.xyxy[0]  # get box coordinates in (left, top, right, bottom)
    c = box.cls
    annotator.box_label(b, model.names[int(c)])

plt.imshow(annotator.result())
```

```
image 1/1 C:\School\yolo5\yolov5\data\images\zidane.jpg: 384x640 (no detectio
156.8ms
Speed: 2.6ms preprocess, 156.8ms inference, 1.0ms postprocess per image at sh
(1, 3, 384, 640)
```

Out[21]: <matplotlib.image.AxesImage at 0x21e8cf15390>

In [4]:
```
from ultralytics import YOLO
#or whatever relative path your folder and files are in
#%cd C:\School\yolo5\yolov5
use_pretrained = True

if use_pretrained:
    print('Using a pretrained model')
    model = YOLO('yolov8n.pt')  # load a pretrained model
else:
    print('Using a scratch model')
    model = YOLO('yolov8n.yaml') # load a scratch model

model.info()

epochs = 1                      # quick test first
data = "lab6coco128.yml"  #

results = model.train(
    data=data,
    epochs=epochs,
    imgsz=640,
```

```
    device=0
)
```

```
C:\Users\nothi\anaconda3\envs\yolov5\lib\site-
packages\IPython\core\magics\osm.py:417: UserWarning: This is now an optional
IPython functionality, setting dhist requires you to install the `pickleshare
library.
  self.shell.db['dhist'] = compress_dhist(dhist)[-100:]
```

```
C:\School\yolo5\yolov5
Using a pretrained model
YOLOv8n summary: 129 layers, 3,157,200 parameters, 0 gradients, 8.9 GFLOPs
Ultralytics 8.4.24  Python-3.10.20 torch-2.7.1+cu118 CUDA:0 (NVIDIA GeForce R
4060 Laptop GPU, 8188MiB)
engine\trainer: agnostic_nms=False, amp=True, angle=1.0, augment=False,
auto_augment=randaugment, batch=16, bgr=0.0, box=7.5, cache=False, cfg=None,
classes=None, close_mosaic=10, cls=0.5, compile=False, conf=None, copy_paste=
copy_paste_mode=flip, cos_lr=False, cutmix=0.0, data=lab6coco128.yml, degrees
deterministic=True, device=0, dfl=1.5, dnn=False, dropout=0.0, dynamic=False,
embed=None, end2end=None, epochs=1, erasing=0.4, exist_ok=False, fliplr=0.5,
flipud=0.0, format=torchscript, fraction=1.0, freeze=None, half=False, hsv_h=
hsv_s=0.7, hsv_v=0.4, imgsz=640, int8=False, iou=0.7, keras=False, kobj=1.0,
line_width=None, lr0=0.01, lrf=0.01, mask_ratio=4, max_det=300, mixup=0.0,
mode=train, model=yolov8n.pt, momentum=0.937, mosaic=1.0, multi_scale=0.0,
name=train, nbs=64, nms=False, opset=None, optimize=False, optimizer=auto,
overlap_mask=True, patience=100, perspective=0.0, plots=True, pose=12.0,
pretrained=True, profile=False, project=None, rect=False, resume=False,
retina_masks=False, rle=1.0, save=True, save_conf=False, save_crop=False,
save_dir=C:\School\yolo5\yolov5\runs\detect\train, save_frames=False,
save_json=False, save_period=-1, save_txt=False, scale=0.5, seed=0, shear=0.0
show=False, show_boxes=True, show_conf=True, show_labels=True, simplify=True,
single_cls=False, source=None, split=val, stream_buffer=False, task=detect,
time=None, tracker=botsort.yaml, translate=0.1, val=True, verbose=True,
vid_stride=1, visualize=False, warmup_bias_lr=0.1, warmup_epochs=3.0,
warmup_momentum=0.8, weight_decay=0.0005, workers=8, workspace=None
Downloading https://ultralytics.com/assets/Arial.ttf to 'C:
\Users\nothi\AppData\Roaming\Ultralytics\Arial.ttf': 100% ━━━━━━━━━━━━ 755.1K
9.0MB/s 0.1s

                   from  n    params  module
arguments
  0                  -1  1       464  ultralytics.nn.modules.conv.Conv
[3, 16, 3, 2]
  1                  -1  1      4672  ultralytics.nn.modules.conv.Conv
[16, 32, 3, 2]
  2                  -1  1      7360  ultralytics.nn.modules.block.C2f
[32, 32, 1, True]
  3                  -1  1     18560  ultralytics.nn.modules.conv.Conv
[32, 64, 3, 2]
  4                  -1  2     49664  ultralytics.nn.modules.block.C2f
[64, 64, 2, True]
  5                  -1  1     73984  ultralytics.nn.modules.conv.Conv
[64, 128, 3, 2]
  6                  -1  2    197632  ultralytics.nn.modules.block.C2f
[128, 128, 2, True]
  7                  -1  1    295424  ultralytics.nn.modules.conv.Conv
[128, 256, 3, 2]
  8                  -1  1    460288  ultralytics.nn.modules.block.C2f
[256, 256, 1, True]
  9                  -1  1    164608  ultralytics.nn.modules.block.SPPF
[256, 256, 5]
 10                  -1  1         0  torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']
 11             [-1, 6]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 12                  -1  1    148224  ultralytics.nn.modules.block.C2f
[384, 128, 1]
 13                  -1  1         0  torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']
```

```
 14             [-1, 4]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 15                  -1  1     37248  ultralytics.nn.modules.block.C2f
[192, 64, 1]
 16                  -1  1     36992  ultralytics.nn.modules.conv.Conv
[64, 64, 3, 2]
 17            [-1, 12]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 18                  -1  1    123648  ultralytics.nn.modules.block.C2f
[192, 128, 1]
 19                  -1  1    147712  ultralytics.nn.modules.conv.Conv
[128, 128, 3, 2]
 20             [-1, 9]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 21                  -1  1    493056  ultralytics.nn.modules.block.C2f
[384, 256, 1]
 22        [15, 18, 21]  1    897664  ultralytics.nn.modules.head.Detect
[80, 16, None, [64, 128, 256]]
Model summary: 130 layers, 3,157,200 parameters, 3,157,184 gradients, 8.9 GFL

Transferred 355/355 items from pretrained weights
Freezing layer 'model.22.dfl.conv.weight'
AMP: running Automatic Mixed Precision (AMP) checks...
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/
yolo26n.pt to 'yolo26n.pt': 100% ━━━━━━━━━━━━ 5.3MB 12.0MB/s 0.4s.4s<0.0s.0s
AMP: checks passed
train: Fast image access  (ping: 0.30.0 ms, read: 61.728.1 MB/s, size: 50.9 K
train: Scanning C:\School\yolo5\datasets\coco128\labels\train2017... 126 imag
backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 128/128 681.3it/s 0.2s3s
train: New cache created: C:\School\yolo5\datasets\coco128\labels\train2017.c
val: Fast image access  (ping: 0.20.0 ms, read: 166.574.0 MB/s, size: 52.5 KB
val: Scanning C:\School\yolo5\datasets\coco128\labels\train2017.cache... 126
images, 2 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 128/128  0.0s
optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and 'momentum=0.937' a
determining best 'optimizer', 'lr0' and 'momentum' automatically...
optimizer: AdamW(lr=0.000119, momentum=0.9) with parameter groups 57
weight(decay=0.0), 64 weight(decay=0.0005), 63 bias(decay=0.0)
Plotting labels to C:\School\yolo5\yolov5\runs\detect\train\labels.jpg...
Image sizes 640 train, 640 val
Using 8 dataloader workers
Logging results to C:\School\yolo5\yolov5\runs\detect\train
Starting training for 1 epochs...

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
        1/1      2.41G      1.268      1.607      1.314        162        640
━━━━━━━━━━━━ 8/8 2.0s/it 15.6s1.0ss
                 Class     Images  Instances      Box(P          R      mAP50
mAP50-95): 100% ━━━━━━━━━━━━ 4/4 1.2s/it 4.8s1.5ss
                   all        128        929      0.646       0.53      0.612
0.451

1 epochs completed in 0.008 hours.
Optimizer stripped from C:\School\yolo5\yolov5\runs\detect\train\weights\last
6.5MB
Optimizer stripped from C:\School\yolo5\yolov5\runs\detect\train\weights\best
6.5MB

Validating C:\School\yolo5\yolov5\runs\detect\train\weights\best.pt...
Ultralytics 8.4.24  Python-3.10.20 torch-2.7.1+cu118 CUDA:0 (NVIDIA GeForce R
4060 Laptop GPU, 8188MiB)
```

```
Model summary (fused): 73 layers, 3,151,904 parameters, 0 gradients, 8.7 GFLO
                 Class     Images  Instances      Box(P          R      mAP50
mAP50-95): 100% ━━━━━━━━━━━━━━━━━━━━ 4/4 1.5it/s 2.6s1.0s
                   all        128        929      0.651      0.532       0.61
0.45
                person         61        254      0.778      0.676      0.761
0.543
               bicycle          3          6      0.517      0.333      0.328
0.27
                   car         12         46      0.811      0.217      0.282
0.175
            motorcycle          4          5      0.603        0.8       0.88
0.679
              airplane          5          6      0.659      0.648      0.783
0.627
                   bus          5          7      0.576      0.714       0.72
0.673
                 train          3          3      0.516      0.667      0.731
0.623
                 truck          5         12          1      0.317      0.487
0.27
                  boat          2          6      0.265      0.167      0.382
0.257
         traffic light          4         14      0.735      0.201      0.206
0.14
             stop sign          2          2      0.843          1      0.995
0.704
                 bench          5          9      0.832      0.552      0.602
0.342
                  bird          2         16      0.919      0.708      0.867
0.527
                   cat          4          4          1      0.956      0.995
0.772
                   dog          9          9      0.576      0.778      0.769
0.567
                 horse          1          2      0.565          1      0.995
0.515
              elephant          4         17      0.944      0.824      0.885
0.654
                  bear          1          1      0.642          1      0.995
0.995
                 zebra          2          4      0.857          1      0.995
0.965
               giraffe          4          9      0.741          1      0.963
0.727
              backpack          4          6      0.644      0.333      0.377
0.235
              umbrella          4         18      0.759      0.444      0.622
0.428
               handbag          9         19      0.606     0.0861      0.206
0.101
                   tie          6          7      0.828      0.691      0.671
0.508
              suitcase          2          4      0.649      0.935      0.828
0.592
               frisbee          5          5      0.669        0.8      0.759
0.655
                  skis          1          1      0.646          1      0.995
0.497
             snowboard          2          7      0.762      0.714      0.768
```

0.475						
	sports ball	6	6	0.585	0.333	0.476
0.269						
	kite	2	10	0.753	0.4	0.507
0.171						
	baseball bat	4	4	0.603	0.5	0.503
0.299						
	baseball glove	4	7	0.655	0.429	0.429
0.295						
	skateboard	3	5	0.849	0.6	0.6
0.44						
	tennis racket	5	7	0.531	0.33	0.49
0.315						
	bottle	6	18	0.429	0.334	0.372
0.226						
	wine glass	5	16	0.823	0.312	0.553
0.334						
	cup	10	36	0.589	0.279	0.409
0.284						
	fork	6	6	0.524	0.167	0.291
0.22						
	knife	7	16	0.667	0.5	0.617
0.37						
	spoon	5	22	0.762	0.182	0.342
0.214						
	bowl	9	28	0.729	0.643	0.627
0.504						
	banana	1	1	0	0	0.0995
0.0398						
	sandwich	2	2	0.248	0.5	0.414
0.414						
	orange	1	4	1	0.35	0.995
0.666						
	broccoli	4	11	0.402	0.182	0.271
0.215						
	carrot	3	24	0.677	0.438	0.589
0.373						
	hot dog	1	2	0.461	1	0.497
0.497						
	pizza	5	5	0.684	1	0.995
0.893						
	donut	2	14	0.664	1	0.914
0.827						
	cake	4	4	0.751	1	0.995
0.88						
	chair	9	35	0.508	0.514	0.454
0.255						
	couch	5	6	0.749	0.498	0.617
0.458						
	potted plant	9	14	0.678	0.714	0.746
0.493						
	bed	3	3	0.643	0.667	0.747
0.581						
	dining table	10	13	0.498	0.615	0.535
0.419						
	toilet	2	2	0.613	0.5	0.828
0.796						
	tv	2	2	0.264	0.5	0.695
0.606						
	laptop	2	3	1	0	0.388

```
0.322
                 mouse          2          2          1          0     0.0488
0.00487
                remote          5          8       0.87        0.5      0.579
0.497
            cell phone          5          8          0          0     0.0451
0.0276
             microwave          3          3       0.49      0.667       0.83
0.733
                  oven          5          5       0.41        0.4       0.34
0.271
                  sink          4          6      0.371      0.167      0.172
0.116
          refrigerator          5          5      0.681        0.4      0.628
0.486
                  book          6         29      0.614      0.111      0.389
0.18
                 clock          8          9      0.783      0.778      0.887
0.767
                  vase          2          2      0.304          1      0.828
0.795
              scissors          1          1          1          0      0.497
0.109
            teddy bear          6         21        0.9       0.43      0.626
0.401
            toothbrush          2          5      0.548      0.258       0.62
0.38
Speed: 0.3ms preprocess, 2.8ms inference, 0.0ms loss, 4.0ms postprocess per i
Results saved to C:\School\yolo5\yolov5\runs\detect\train
```

In [5]:
```
#%cd C:\School\yolo5\ or whatpath you like
```

```
C:\School\yolo5
```

```
C:\Users\nothi\anaconda3\envs\yolov5\lib\site-
packages\IPython\core\magics\osm.py:417: UserWarning: This is now an optional
IPython functionality, setting dhist requires you to install the `pickleshare
library.
  self.shell.db['dhist'] = compress_dhist(dhist)[-100:]
```

In [6]:
```
import os
import xml.etree.ElementTree as ET
from pathlib import Path

# Function to get the data from XML Annotation
def extract_info_from_xml(xml_file):
    root = ET.parse(xml_file).getroot()

    info_dict = {}
    info_dict["bboxes"] = []

    for elem in root:
        if elem.tag == "filename":
            info_dict["filename"] = elem.text

        elif elem.tag == "size":
            image_size = []
            for subelem in elem:
                image_size.append(int(subelem.text))
            info_dict["image_size"] = tuple(image_size)  # (w, h, c)
```

```
        elif elem.tag == "object":
            bbox = {}
            for subelem in elem:
                if subelem.tag == "name":
                    bbox["class"] = subelem.text

                elif subelem.tag == "bndbox":
                    for subsubelem in subelem:
                        bbox[subsubelem.tag] = int(subsubelem.text)

            info_dict["bboxes"].append(bbox)

    return info_dict


class_name_to_id_mapping = {
    "trafficlight": 0,
    "stop": 1,
    "speedlimit": 2,
    "crosswalk": 3
}


def convert_to_yolo(info_dict, label_output_dir, write_to_file=True):
    lines = []

    for b in info_dict["bboxes"]:
        if b["class"] not in class_name_to_id_mapping:
            print("Skipping invalid class:", b["class"])
            continue

        class_id = class_name_to_id_mapping[b["class"]]

        # YOLO box format
        b_center_x = (b["xmin"] + b["xmax"]) / 2
        b_center_y = (b["ymin"] + b["ymax"]) / 2
        b_width = b["xmax"] - b["xmin"]
        b_height = b["ymax"] - b["ymin"]

        image_w, image_h, image_c = info_dict["image_size"]

        # normalize
        b_center_x /= image_w
        b_center_y /= image_h
        b_width /= image_w
        b_height /= image_h

        lines.append(f"{class_id} {b_center_x:.6f} {b_center_y:.6f} {b_wi

    if write_to_file:
        os.makedirs(label_output_dir, exist_ok=True)
        txt_name = Path(info_dict["filename"]).stem + ".txt"
        save_file_name = os.path.join(label_output_dir, txt_name)

        with open(save_file_name, "w", encoding="utf-8") as f:
            f.write("\n".join(lines))
    else:
        return lines
```

In [7]:
```
from pathlib import Path

import os

print(os.getcwd())
print(os.listdir("road-sign-detection"))
```

```
C:\School\yolo5
['annotations', 'images', 'Readme.md']
```

In [8]:
```
#from pathlib import Path

root = Path("road-sign-detection")
xml_dir = root / "annotations"

one_xml = next(xml_dir.glob("*.xml"))

info = extract_info_from_xml(one_xml)
print(one_xml.name)
print(info["filename"])
print(info["image_size"])
print(info["bboxes"][:2])
```

```
road0.xml
road0.png
(267, 400, 3)
[{'class': 'trafficlight', 'xmin': 98, 'ymin': 62, 'xmax': 208, 'ymax': 232}]
```

In [9]:
```
convert_to_yolo(
    info,
    label_output_dir=root / "labels" / "test_one",
    write_to_file=True
)
```

In [10]:
```
print(list((root / "labels" / "test_one").glob("*.txt"))[:5])
```

```
[WindowsPath('road-sign-detection/labels/test_one/road0.txt')]
```

In [11]:
```
#now convert all
#from pathlib import Path

root = Path("road-sign-detection")
xml_dir = root / "annotations"
label_dir = root / "labels" / "all"

for xml_file in xml_dir.glob("*.xml"):
    info = extract_info_from_xml(xml_file)
    convert_to_yolo(info, label_output_dir=label_dir, write_to_file=True)

print("Done converting XML -> YOLO txt")
print("Number of txt files:", len(list(label_dir.glob("*.txt"))))
```

```
Done converting XML -> YOLO txt
Number of txt files: 877
```

In [12]:
```
import random
import shutil
from pathlib import Path

random.seed(42)
```

```
root = Path("road-sign-detection")
img_src = root / "images"
lbl_src = root / "labels" / "all"

all_images = []
for ext in ["*.png", "*.jpg", "*.jpeg", "*.PNG", "*.JPG", "*.JPEG"]:
    all_images.extend(img_src.glob(ext))

all_images = sorted(all_images)
random.shuffle(all_images)

n = len(all_images)
n_train = int(0.7 * n)
n_val = int(0.15 * n)

train_imgs = all_images[:n_train]
val_imgs = all_images[n_train:n_train + n_val]
test_imgs = all_images[n_train + n_val:]

splits = {
    "train": train_imgs,
    "val": val_imgs,
    "test": test_imgs,
}

for split in ["train", "val", "test"]:
    (root / "images" / split).mkdir(parents=True, exist_ok=True)
    (root / "labels" / split).mkdir(parents=True, exist_ok=True)

for split, files in splits.items():
    for img_path in files:
        shutil.copy2(img_path, root / "images" / split / img_path.name)

        txt_path = lbl_src / f"{img_path.stem}.txt"
        if txt_path.exists():
            shutil.copy2(txt_path, root / "labels" / split / txt_path.nam
        else:
            print("Missing label for:", img_path.name)

print("Done splitting dataset")
print("train:", len(train_imgs), "val:", len(val_imgs), "test:", len(test
```

```
Done splitting dataset
train: 1227 val: 263 test: 264
```

In [13]:

```
#from pathlib import Path

root = Path("road-sign-detection")

for split in ["train", "val", "test"]:
    n_imgs = len(list((root / "images" / split).glob("*")))
    n_lbls = len(list((root / "labels" / split).glob("*.txt")))
    print(split, "images:", n_imgs, "labels:", n_lbls)
```

```
train images: 787 labels: 787
val images: 239 labels: 239
test images: 243 labels: 243
```

In [14]:

```
yaml_text = """path: road-sign-detection
train: images/train
val: images/val
```

```
test: images/test

nc: 4
names: ["trafficlight", "stop", "speedlimit", "crosswalk"]
"""

with open("roadsigns.yaml", "w", encoding="utf-8") as f:
    f.write(yaml_text)

print("roadsigns.yaml written")
```

```
roadsigns.yaml written
```

In [15]:

```
#Construct a pretrained YOLO and capture its performance on the test set
from ultralytics import YOLO

model = YOLO("yolov8n.pt")   # pretrained tiny/nano model
metrics = model.val(
    data="roadsigns.yaml",
    split="test",
    imgsz=640,
    device=0,
    name="roadsigns_pretrained_testonly"
)

print("mAP50-95 =", metrics.box.map)
print("mAP50    =", metrics.box.map50)
print("mAP75    =", metrics.box.map75)
```

```
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/
yolov8n.pt to 'yolov8n.pt': 100% ━━━━━━━━━━━━ 6.2MB 12.2MB/s 0.5s.4s<0.1s.6s
Ultralytics 8.4.24  Python-3.10.20 torch-2.7.1+cu118 CUDA:0 (NVIDIA GeForce R
4060 Laptop GPU, 8188MiB)
YOLOv8n summary (fused): 72 layers, 3,151,904 parameters, 0 gradients, 8.7 GF
val: Fast image access  (ping: 0.20.0 ms, read: 116.19.7 MB/s, size: 201.7 KB
val: Scanning C:\School\yolo5\road-sign-detection\labels\test... 243 images,
backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 243/243 390.1it/s 0.6s0.1s
val: New cache created: C:\School\yolo5\road-sign-detection\labels\test.cache
                 Class     Images  Instances      Box(P          R      mAP50
mAP50-95): 100% ━━━━━━━━━━━━ 16/16 3.7it/s 4.3s0.2s
                   all        243        343   0.000397     0.0266   0.000226
8.26e-05
                person         32         47    0.00159      0.106   0.000904
0.000331
               bicycle         29         29          0          0
0          0
                   car        188        218          0          0
0          0
            motorcycle         45         49          0          0
0          0
Speed: 2.9ms preprocess, 6.4ms inference, 0.0ms loss, 2.2ms postprocess per i
Results saved to C:\School\yolo5\runs\detect\roadsigns_pretrained_testonly
mAP50-95 = 8.263411582844934e-05
mAP50    = 0.00022598045805112996
mAP75    = 4.2602266410713175e-05
```

In [3]:

```
"""The pretrained YOLOv8n model achieved extremely poor performance on th
This is expected because the pretrained model is trained on COCO classes,
while the target dataset uses custom road-sign classes.
Therefore, before training, the model class semantics do not align with t
mAP50    = 0.00022598045805112996
```

```
mAP75    = 4.2602266410713175e-05
"""
# step 2 Construct a pretrained YOLO. Train it for 5 epochs on the traini
from ultralytics import YOLO
%cd C:\School\yolo5
model = YOLO("yolov8n.pt")

train_results = model.train(
    data="roadsigns.yaml",
    epochs=5,
    imgsz=640,
    batch=8,
    device=0,
    name="roadsigns_pretrained_5ep_bs8"
)
```

```
C:\Users\nothi\anaconda3\envs\yolov5\lib\site-
packages\IPython\core\magics\osm.py:417: UserWarning: This is now an optional
IPython functionality, setting dhist requires you to install the `pickleshare
library.
  self.shell.db['dhist'] = compress_dhist(dhist)[-100:]
```

```
C:\School\yolo5
Ultralytics 8.4.24  Python-3.10.20 torch-2.7.1+cu118 CUDA:0 (NVIDIA GeForce R
4060 Laptop GPU, 8188MiB)
engine\trainer: agnostic_nms=False, amp=True, angle=1.0, augment=False,
auto_augment=randaugment, batch=8, bgr=0.0, box=7.5, cache=False, cfg=None,
classes=None, close_mosaic=10, cls=0.5, compile=False, conf=None, copy_paste=
copy_paste_mode=flip, cos_lr=False, cutmix=0.0, data=roadsigns.yaml, degrees=
deterministic=True, device=0, dfl=1.5, dnn=False, dropout=0.0, dynamic=False,
embed=None, end2end=None, epochs=5, erasing=0.4, exist_ok=False, fliplr=0.5,
flipud=0.0, format=torchscript, fraction=1.0, freeze=None, half=False, hsv_h=
hsv_s=0.7, hsv_v=0.4, imgsz=640, int8=False, iou=0.7, keras=False, kobj=1.0,
line_width=None, lr0=0.01, lrf=0.01, mask_ratio=4, max_det=300, mixup=0.0,
mode=train, model=yolov8n.pt, momentum=0.937, mosaic=1.0, multi_scale=0.0,
name=roadsigns_pretrained_5ep_bs8, nbs=64, nms=False, opset=None, optimize=Fa
optimizer=auto, overlap_mask=True, patience=100, perspective=0.0, plots=True,
pose=12.0, pretrained=True, profile=False, project=None, rect=False, resume=F
retina_masks=False, rle=1.0, save=True, save_conf=False, save_crop=False,
save_dir=C:\School\yolo5\runs\detect\roadsigns_pretrained_5ep_bs8,
save_frames=False, save_json=False, save_period=-1, save_txt=False, scale=0.5
seed=0, shear=0.0, show=False, show_boxes=True, show_conf=True, show_labels=T
simplify=True, single_cls=False, source=None, split=val, stream_buffer=False,
task=detect, time=None, tracker=botsort.yaml, translate=0.1, val=True,
verbose=True, vid_stride=1, visualize=False, warmup_bias_lr=0.1, warmup_epoch
warmup_momentum=0.8, weight_decay=0.0005, workers=8, workspace=None
Overriding model.yaml nc=80 with nc=4

                   from  n    params  module
arguments
  0                  -1  1       464  ultralytics.nn.modules.conv.Conv
[3, 16, 3, 2]
  1                  -1  1      4672  ultralytics.nn.modules.conv.Conv
[16, 32, 3, 2]
  2                  -1  1      7360  ultralytics.nn.modules.block.C2f
[32, 32, 1, True]
  3                  -1  1     18560  ultralytics.nn.modules.conv.Conv
[32, 64, 3, 2]
  4                  -1  2     49664  ultralytics.nn.modules.block.C2f
[64, 64, 2, True]
  5                  -1  1     73984  ultralytics.nn.modules.conv.Conv
[64, 128, 3, 2]
  6                  -1  2    197632  ultralytics.nn.modules.block.C2f
[128, 128, 2, True]
  7                  -1  1    295424  ultralytics.nn.modules.conv.Conv
[128, 256, 3, 2]
  8                  -1  1    460288  ultralytics.nn.modules.block.C2f
[256, 256, 1, True]
  9                  -1  1    164608  ultralytics.nn.modules.block.SPPF
[256, 256, 5]
 10                  -1  1         0  torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']
 11             [-1, 6]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 12                  -1  1    148224  ultralytics.nn.modules.block.C2f
[384, 128, 1]
 13                  -1  1         0  torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']
 14             [-1, 4]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 15                  -1  1     37248  ultralytics.nn.modules.block.C2f
[192, 64, 1]
```

```
 16                  -1  1     36992  ultralytics.nn.modules.conv.Conv
[64, 64, 3, 2]
 17            [-1, 12]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 18                  -1  1    123648  ultralytics.nn.modules.block.C2f
[192, 128, 1]
 19                  -1  1    147712  ultralytics.nn.modules.conv.Conv
[128, 128, 3, 2]
 20             [-1, 9]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 21                  -1  1    493056  ultralytics.nn.modules.block.C2f
[384, 256, 1]
 22        [15, 18, 21]  1    752092  ultralytics.nn.modules.head.Detect
[4, 16, None, [64, 128, 256]]
Model summary: 130 layers, 3,011,628 parameters, 3,011,612 gradients, 8.2 GFL

Transferred 319/355 items from pretrained weights
Freezing layer 'model.22.dfl.conv.weight'
AMP: running Automatic Mixed Precision (AMP) checks...
AMP: checks passed
train: Fast image access  (ping: 0.10.0 ms, read: 380.0109.6 MB/s, size: 325.
train: Scanning C:\School\yolo5\road-sign-detection\labels\train.cache... 787
images, 0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 787/787  0.0s
val: Fast image access  (ping: 0.50.2 ms, read: 98.155.9 MB/s, size: 264.5 KB
val:
Scanning C:\School\yolo5\road-sign-detection\labels\val.cache... 239 images,
backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 239/239  0.0s
optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and 'momentum=0.937' a
determining best 'optimizer', 'lr0' and 'momentum' automatically...
optimizer: AdamW(lr=0.00125, momentum=0.9) with parameter groups 57
weight(decay=0.0), 64 weight(decay=0.0005), 63 bias(decay=0.0)
Plotting labels to C:
\School\yolo5\runs\detect\roadsigns_pretrained_5ep_bs8\labels.jpg...
Image sizes 640 train, 640 val
Using 8 dataloader workers
Logging results to C:\School\yolo5\runs\detect\roadsigns_pretrained_5ep_bs8
Starting training for 5 epochs...

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
        1/5      1.08G     0.7947      2.263     0.9469          7        640
━━━━━━━━━━━━ 99/99 4.2it/s 23.7s0.2ss
                 Class     Images  Instances      Box(P          R      mAP50
mAP50-95): 100% ━━━━━━━━━━━━ 15/15 4.5it/s 3.3s0.2s
                   all        239        343      0.855      0.443       0.75
0.564

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
        2/5      1.24G     0.7704      1.375     0.9506          8        640
━━━━━━━━━━━━ 99/99 5.3it/s 18.8s0.2s
                 Class     Images  Instances      Box(P          R      mAP50
mAP50-95): 100% ━━━━━━━━━━━━ 15/15 5.6it/s 2.7s0.2s
                   all        239        343      0.902      0.781      0.859
0.645

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
        3/5      1.27G     0.7522      1.192     0.9468          8        640
━━━━━━━━━━━━ 99/99 5.4it/s 18.2s0.2s
                 Class     Images  Instances      Box(P          R      mAP50
mAP50-95): 100% ━━━━━━━━━━━━ 15/15 5.5it/s 2.8s0.2s
                   all        239        343      0.856      0.845      0.899
```

```
0.715

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
        4/5      1.27G     0.7214      1.052     0.9247         11        640
━━━━━━━━━━━━ 99/99 5.5it/s 18.1s0.2s
                 Class     Images  Instances      Box(P          R      mAP50
mAP50-95): 100% ━━━━━━━━━━━━ 15/15 5.6it/s 2.7s0.2s
                   all        239        343      0.914      0.883      0.921
0.742

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
        5/5       1.3G     0.6585     0.9333     0.9214         10        640
━━━━━━━━━━━━ 99/99 5.4it/s 18.2s0.2s
                 Class     Images  Instances      Box(P          R      mAP50
mAP50-95): 100% ━━━━━━━━━━━━ 15/15 5.6it/s 2.7s0.2s
                   all        239        343      0.928      0.928      0.954
0.789

5 epochs completed in 0.032 hours.
Optimizer stripped from C:
\School\yolo5\runs\detect\roadsigns_pretrained_5ep_bs8\weights\last.pt, 6.2MB
Optimizer stripped from C:
\School\yolo5\runs\detect\roadsigns_pretrained_5ep_bs8\weights\best.pt, 6.2MB

Validating C:
\School\yolo5\runs\detect\roadsigns_pretrained_5ep_bs8\weights\best.pt...
Ultralytics 8.4.24  Python-3.10.20 torch-2.7.1+cu118 CUDA:0 (NVIDIA GeForce R
4060 Laptop GPU, 8188MiB)
Model summary (fused): 73 layers, 3,006,428 parameters, 0 gradients, 8.1 GFLO
                 Class     Images  Instances      Box(P          R      mAP50
mAP50-95): 100% ━━━━━━━━━━━━ 15/15 4.6it/s 3.2s0.2s
                   all        239        343      0.928      0.928      0.954
0.79
          trafficlight         25         37      0.832      0.838      0.886
0.598
                  stop         27         27      0.981      0.963      0.991
0.932
            speedlimit        182        212      0.988      0.995      0.995
0.887
             crosswalk         53         67      0.911      0.917      0.944
0.742
Speed: 0.3ms preprocess, 2.6ms inference, 0.0ms loss, 2.8ms postprocess per i
Results saved to C:\School\yolo5\runs\detect\roadsigns_pretrained_5ep_bs8
```

In [1]:
```
import torch
torch.cuda.empty_cache()
```

In [4]:
```
#evaluation
from ultralytics import YOLO

best_model = YOLO("runs/detect/roadsigns_pretrained_5ep_bs8/weights/best.

metrics = best_model.val(
    data="roadsigns.yaml",
    split="test",
    imgsz=640,
    device=0,
    name="roadsigns_pretrained_5ep_bs8_test"
)
```

```
print("P        =", metrics.box.mp)
print("R        =", metrics.box.mr)
print("mAP50    =", metrics.box.map50)
print("mAP50-95 =", metrics.box.map)
```

```
Ultralytics 8.4.24  Python-3.10.20 torch-2.7.1+cu118 CUDA:0 (NVIDIA GeForce R
4060 Laptop GPU, 8188MiB)
Model summary (fused): 73 layers, 3,006,428 parameters, 0 gradients, 8.1 GFLO
val: Fast image access  (ping: 0.60.1 ms, read: 216.272.3 MB/s, size: 260.6 K
val: Scanning C:\School\yolo5\road-sign-detection\labels\test.cache... 243 im
0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 243/243  0.0s
                 Class     Images  Instances      Box(P          R      mAP50
mAP50-95): 100% ━━━━━━━━━━━━ 16/16 3.0it/s 5.4s0.2s
                   all        243        343      0.938      0.889      0.928
0.773
          trafficlight         32         47      0.872      0.723      0.835
0.568
                  stop         29         29          1      0.983      0.995
0.925
            speedlimit        188        218      0.981      0.991      0.988
0.884
             crosswalk         45         49      0.899      0.857      0.894
0.718
Speed: 5.3ms preprocess, 6.0ms inference, 0.0ms loss, 2.4ms postprocess per i
Results saved to C:\School\yolo5\runs\detect\roadsigns_pretrained_5ep_bs8_tes
P        = 0.9376656281285847
R        = 0.8886384109120996
mAP50    = 0.9281838773251658
mAP50-95 = 0.7734831843829346
```

In [5]:

```
from ultralytics import YOLO
#step3 scratch model 5
model = YOLO("yolov8n.yaml")

train_results = model.train(
    data="roadsigns.yaml",
    epochs=5,
    imgsz=640,
    batch=8,
    device=0,
    name="roadsigns_scratch_5ep_bs8"
)
```

```
Ultralytics 8.4.24  Python-3.10.20 torch-2.7.1+cu118 CUDA:0 (NVIDIA GeForce R
4060 Laptop GPU, 8188MiB)
engine\trainer: agnostic_nms=False, amp=True, angle=1.0, augment=False,
auto_augment=randaugment, batch=8, bgr=0.0, box=7.5, cache=False, cfg=None,
classes=None, close_mosaic=10, cls=0.5, compile=False, conf=None, copy_paste=
copy_paste_mode=flip, cos_lr=False, cutmix=0.0, data=roadsigns.yaml, degrees=
deterministic=True, device=0, dfl=1.5, dnn=False, dropout=0.0, dynamic=False,
embed=None, end2end=None, epochs=5, erasing=0.4, exist_ok=False, fliplr=0.5,
flipud=0.0, format=torchscript, fraction=1.0, freeze=None, half=False, hsv_h=
hsv_s=0.7, hsv_v=0.4, imgsz=640, int8=False, iou=0.7, keras=False, kobj=1.0,
line_width=None, lr0=0.01, lrf=0.01, mask_ratio=4, max_det=300, mixup=0.0,
mode=train, model=yolov8n.yaml, momentum=0.937, mosaic=1.0, multi_scale=0.0,
name=roadsigns_scratch_5ep_bs8, nbs=64, nms=False, opset=None, optimize=False
optimizer=auto, overlap_mask=True, patience=100, perspective=0.0, plots=True,
pose=12.0, pretrained=True, profile=False, project=None, rect=False, resume=F
retina_masks=False, rle=1.0, save=True, save_conf=False, save_crop=False,
save_dir=C:\School\yolo5\runs\detect\roadsigns_scratch_5ep_bs8, save_frames=F
save_json=False, save_period=-1, save_txt=False, scale=0.5, seed=0, shear=0.0
show=False, show_boxes=True, show_conf=True, show_labels=True, simplify=True,
single_cls=False, source=None, split=val, stream_buffer=False, task=detect,
time=None, tracker=botsort.yaml, translate=0.1, val=True, verbose=True,
vid_stride=1, visualize=False, warmup_bias_lr=0.1, warmup_epochs=3.0,
warmup_momentum=0.8, weight_decay=0.0005, workers=8, workspace=None
Overriding model.yaml nc=80 with nc=4

                   from  n    params  module
arguments
  0                  -1  1       464  ultralytics.nn.modules.conv.Conv
[3, 16, 3, 2]
  1                  -1  1      4672  ultralytics.nn.modules.conv.Conv
[16, 32, 3, 2]
  2                  -1  1      7360  ultralytics.nn.modules.block.C2f
[32, 32, 1, True]
  3                  -1  1     18560  ultralytics.nn.modules.conv.Conv
[32, 64, 3, 2]
  4                  -1  2     49664  ultralytics.nn.modules.block.C2f
[64, 64, 2, True]
  5                  -1  1     73984  ultralytics.nn.modules.conv.Conv
[64, 128, 3, 2]
  6                  -1  2    197632  ultralytics.nn.modules.block.C2f
[128, 128, 2, True]
  7                  -1  1    295424  ultralytics.nn.modules.conv.Conv
[128, 256, 3, 2]
  8                  -1  1    460288  ultralytics.nn.modules.block.C2f
[256, 256, 1, True]
  9                  -1  1    164608  ultralytics.nn.modules.block.SPPF
[256, 256, 5]
 10                  -1  1         0  torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']
 11             [-1, 6]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 12                  -1  1    148224  ultralytics.nn.modules.block.C2f
[384, 128, 1]
 13                  -1  1         0  torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']
 14             [-1, 4]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 15                  -1  1     37248  ultralytics.nn.modules.block.C2f
[192, 64, 1]
 16                  -1  1     36992  ultralytics.nn.modules.conv.Conv
```

```
[64, 64, 3, 2]
 17            [-1, 12]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 18                  -1  1    123648  ultralytics.nn.modules.block.C2f
[192, 128, 1]
 19                  -1  1    147712  ultralytics.nn.modules.conv.Conv
[128, 128, 3, 2]
 20             [-1, 9]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 21                  -1  1    493056  ultralytics.nn.modules.block.C2f
[384, 256, 1]
 22        [15, 18, 21]  1    752092  ultralytics.nn.modules.head.Detect
[4, 16, None, [64, 128, 256]]
YOLOv8n summary: 130 layers, 3,011,628 parameters, 3,011,612 gradients, 8.2 G

Freezing layer 'model.22.dfl.conv.weight'
AMP: running Automatic Mixed Precision (AMP) checks...
AMP: checks passed
train: Fast image access  (ping: 0.30.1 ms, read: 234.062.4 MB/s, size: 325.0
train: Scanning C:\School\yolo5\road-sign-detection\labels\train.cache... 787
images, 0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 787/787 875.6Kit/s 0.0s
val: Fast image access  (ping: 0.70.9 ms, read: 190.6147.2 MB/s, size: 264.5
val:
Scanning C:\School\yolo5\road-sign-detection\labels\val.cache... 239 images,
backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 239/239  0.0s
optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and 'momentum=0.937' a
determining best 'optimizer', 'lr0' and 'momentum' automatically...
optimizer: AdamW(lr=0.00125, momentum=0.9) with parameter groups 57
weight(decay=0.0), 64 weight(decay=0.0005), 63 bias(decay=0.0)
Plotting labels to C:
\School\yolo5\runs\detect\roadsigns_scratch_5ep_bs8\labels.jpg...
Image sizes 640 train, 640 val
Using 8 dataloader workers
Logging results to C:\School\yolo5\runs\detect\roadsigns_scratch_5ep_bs8
Starting training for 5 epochs...

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
        1/5      1.08G       3.59      5.418      4.275          7        640
━━━━━━━━━━━━ 99/99 4.5it/s 22.1s0.2ss
                 Class     Images  Instances      Box(P          R      mAP50
mAP50-95): 100% ━━━━━━━━━━━━ 15/15 5.7it/s 2.6s0.2s
                   all        239        343   5.71e-05     0.0116   2.94e-05
1.26e-05

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
        2/5      1.09G      3.167      4.656      3.708          8        640
━━━━━━━━━━━━ 99/99 5.3it/s 18.7s0.2s
                 Class     Images  Instances      Box(P          R      mAP50
mAP50-95): 100% ━━━━━━━━━━━━ 15/15 5.5it/s 2.7s0.2s
                   all        239        343     0.0181     0.0665    0.00888
0.00329

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
        3/5      1.09G      2.815      3.995      3.169          8        640
━━━━━━━━━━━━ 99/99 5.3it/s 18.7s0.2s
                 Class     Images  Instances      Box(P          R      mAP50
mAP50-95): 100% ━━━━━━━━━━━━ 15/15 5.6it/s 2.7s0.2s
                   all        239        343      0.324      0.169     0.0541
0.0238
```

```
      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
        4/5      1.09G      2.464      3.329      2.727         11        640
━━━━━━━━━━━━ 99/99 5.6it/s 17.8s0.2s
                 Class     Images  Instances      Box(P          R      mAP50
mAP50-95): 100% ━━━━━━━━━━━━ 15/15 5.6it/s 2.7s0.2s
                   all        239        343       0.45      0.247      0.243
0.139

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
        5/5      1.09G      2.233      2.837      2.561         10        640
━━━━━━━━━━━━ 99/99 5.5it/s 17.9s0.2s
                 Class     Images  Instances      Box(P          R      mAP50
mAP50-95): 100% ━━━━━━━━━━━━ 15/15 5.6it/s 2.7s0.2s
                   all        239        343      0.654      0.306      0.329
0.193

5 epochs completed in 0.032 hours.
Optimizer stripped from C:
\School\yolo5\runs\detect\roadsigns_scratch_5ep_bs8\weights\last.pt, 6.2MB
Optimizer stripped from C:
\School\yolo5\runs\detect\roadsigns_scratch_5ep_bs8\weights\best.pt, 6.2MB

Validating C:\School\yolo5\runs\detect\roadsigns_scratch_5ep_bs8\weights\best
Ultralytics 8.4.24  Python-3.10.20 torch-2.7.1+cu118 CUDA:0 (NVIDIA GeForce R
4060 Laptop GPU, 8188MiB)
YOLOv8n summary (fused): 73 layers, 3,006,428 parameters, 0 gradients, 8.1 GF
                 Class     Images  Instances      Box(P          R      mAP50
mAP50-95): 100% ━━━━━━━━━━━━ 15/15 4.7it/s 3.2s0.2s
                   all        239        343       0.65      0.304      0.329
0.193
          trafficlight         25         37          1          0     0.0338
0.0106
                  stop         27         27      0.703      0.185      0.332
0.154
            speedlimit        182        212      0.506      0.675      0.673
0.458
             crosswalk         53         67      0.391      0.358      0.276
0.152
Speed: 0.4ms preprocess, 3.2ms inference, 0.0ms loss, 3.0ms postprocess per i
Results saved to C:\School\yolo5\runs\detect\roadsigns_scratch_5ep_bs8
```

In [6]:

```python
#test after 5 stratch train
best_model = YOLO("runs/detect/roadsigns_scratch_5ep_bs8/weights/best.pt"

metrics = best_model.val(
    data="roadsigns.yaml",
    split="test",
    imgsz=640,
    device=0,
    name="roadsigns_scratch_5ep_bs8_test"
)

print("P        =", metrics.box.mp)
print("R        =", metrics.box.mr)
print("mAP50    =", metrics.box.map50)
print("mAP50-95 =", metrics.box.map)
```

```
Ultralytics 8.4.24  Python-3.10.20 torch-2.7.1+cu118 CUDA:0 (NVIDIA GeForce R
4060 Laptop GPU, 8188MiB)
YOLOv8n summary (fused): 73 layers, 3,006,428 parameters, 0 gradients, 8.1 GF
val: Fast image access  (ping: 0.30.1 ms, read: 225.099.9 MB/s, size: 260.6 K
val: Scanning C:\School\yolo5\road-sign-detection\labels\test.cache... 243 im
0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 243/243  0.0s
                 Class     Images  Instances      Box(P          R      mAP50
mAP50-95): 100% ━━━━━━━━━━━━ 16/16 3.0it/s 5.4s0.2s
                   all        243        343      0.695      0.327      0.363
0.217
          trafficlight         32         47          1          0      0.026
0.00569
                  stop         29         29      0.856      0.205      0.422
0.235
            speedlimit        188        218      0.519      0.716      0.671
0.469
             crosswalk         45         49      0.403      0.388      0.332
0.158
Speed: 6.8ms preprocess, 4.2ms inference, 0.0ms loss, 2.5ms postprocess per i
Results saved to C:\School\yolo5\runs\detect\roadsigns_scratch_5ep_bs8_test
P        = 0.6945996928166346
R        = 0.32718473573367546
mAP50    = 0.3625400032094348
mAP50-95 = 0.21687417707223994
```

In [7]:

```python
from ultralytics import YOLO
#step3 scratch model 15
model = YOLO("yolov8n.yaml")

train_results = model.train(
    data="roadsigns.yaml",
    epochs=15,
    imgsz=640,
    batch=8,
    device=0,
    name="roadsigns_scratch_15ep_bs8"
)
```

```
Ultralytics 8.4.24  Python-3.10.20 torch-2.7.1+cu118 CUDA:0 (NVIDIA GeForce R
4060 Laptop GPU, 8188MiB)
engine\trainer: agnostic_nms=False, amp=True, angle=1.0, augment=False,
auto_augment=randaugment, batch=8, bgr=0.0, box=7.5, cache=False, cfg=None,
classes=None, close_mosaic=10, cls=0.5, compile=False, conf=None, copy_paste=
copy_paste_mode=flip, cos_lr=False, cutmix=0.0, data=roadsigns.yaml, degrees=
deterministic=True, device=0, dfl=1.5, dnn=False, dropout=0.0, dynamic=False,
embed=None, end2end=None, epochs=15, erasing=0.4, exist_ok=False, fliplr=0.5,
flipud=0.0, format=torchscript, fraction=1.0, freeze=None, half=False, hsv_h=
hsv_s=0.7, hsv_v=0.4, imgsz=640, int8=False, iou=0.7, keras=False, kobj=1.0,
line_width=None, lr0=0.01, lrf=0.01, mask_ratio=4, max_det=300, mixup=0.0,
mode=train, model=yolov8n.yaml, momentum=0.937, mosaic=1.0, multi_scale=0.0,
name=roadsigns_scratch_5ep_bs82, nbs=64, nms=False, opset=None, optimize=Fals
optimizer=auto, overlap_mask=True, patience=100, perspective=0.0, plots=True,
pose=12.0, pretrained=True, profile=False, project=None, rect=False, resume=F
retina_masks=False, rle=1.0, save=True, save_conf=False, save_crop=False,
save_dir=C:\School\yolo5\runs\detect\roadsigns_scratch_5ep_bs82, save_frames=
save_json=False, save_period=-1, save_txt=False, scale=0.5, seed=0, shear=0.0
show=False, show_boxes=True, show_conf=True, show_labels=True, simplify=True,
single_cls=False, source=None, split=val, stream_buffer=False, task=detect,
time=None, tracker=botsort.yaml, translate=0.1, val=True, verbose=True,
vid_stride=1, visualize=False, warmup_bias_lr=0.1, warmup_epochs=3.0,
warmup_momentum=0.8, weight_decay=0.0005, workers=8, workspace=None
Overriding model.yaml nc=80 with nc=4

                   from  n    params  module
arguments
  0                  -1  1       464  ultralytics.nn.modules.conv.Conv
[3, 16, 3, 2]
  1                  -1  1      4672  ultralytics.nn.modules.conv.Conv
[16, 32, 3, 2]
  2                  -1  1      7360  ultralytics.nn.modules.block.C2f
[32, 32, 1, True]
  3                  -1  1     18560  ultralytics.nn.modules.conv.Conv
[32, 64, 3, 2]
  4                  -1  2     49664  ultralytics.nn.modules.block.C2f
[64, 64, 2, True]
  5                  -1  1     73984  ultralytics.nn.modules.conv.Conv
[64, 128, 3, 2]
  6                  -1  2    197632  ultralytics.nn.modules.block.C2f
[128, 128, 2, True]
  7                  -1  1    295424  ultralytics.nn.modules.conv.Conv
[128, 256, 3, 2]
  8                  -1  1    460288  ultralytics.nn.modules.block.C2f
[256, 256, 1, True]
  9                  -1  1    164608  ultralytics.nn.modules.block.SPPF
[256, 256, 5]
 10                  -1  1         0  torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']
 11             [-1, 6]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 12                  -1  1    148224  ultralytics.nn.modules.block.C2f
[384, 128, 1]
 13                  -1  1         0  torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']
 14             [-1, 4]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 15                  -1  1     37248  ultralytics.nn.modules.block.C2f
[192, 64, 1]
 16                  -1  1     36992  ultralytics.nn.modules.conv.Conv
```

```
[64, 64, 3, 2]
 17            [-1, 12]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 18                  -1  1    123648  ultralytics.nn.modules.block.C2f
[192, 128, 1]
 19                  -1  1    147712  ultralytics.nn.modules.conv.Conv
[128, 128, 3, 2]
 20             [-1, 9]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 21                  -1  1    493056  ultralytics.nn.modules.block.C2f
[384, 256, 1]
 22        [15, 18, 21]  1    752092  ultralytics.nn.modules.head.Detect
[4, 16, None, [64, 128, 256]]
YOLOv8n summary: 130 layers, 3,011,628 parameters, 3,011,612 gradients, 8.2 G

Freezing layer 'model.22.dfl.conv.weight'
AMP: running Automatic Mixed Precision (AMP) checks...
AMP: checks passed
train: Fast image access  (ping: 0.10.0 ms, read: 340.0103.6 MB/s, size: 325.
train: Scanning C:\School\yolo5\road-sign-detection\labels\train.cache... 787
images, 0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 787/787  0.0s
val: Fast image access  (ping: 0.61.0 ms, read: 189.8157.3 MB/s, size: 264.5
val:
Scanning C:\School\yolo5\road-sign-detection\labels\val.cache... 239 images,
backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 239/239  0.0s
optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and 'momentum=0.937' a
determining best 'optimizer', 'lr0' and 'momentum' automatically...
optimizer: AdamW(lr=0.00125, momentum=0.9) with parameter groups 57
weight(decay=0.0), 64 weight(decay=0.0005), 63 bias(decay=0.0)
Plotting labels to C:
\School\yolo5\runs\detect\roadsigns_scratch_5ep_bs82\labels.jpg...
Image sizes 640 train, 640 val
Using 8 dataloader workers
Logging results to C:\School\yolo5\runs\detect\roadsigns_scratch_5ep_bs82
Starting training for 15 epochs...

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       1/15       1.1G       3.59      5.418      4.275          7        640
━━━━━━━━━━━━ 99/99 4.6it/s 21.7s0.2ss
                 Class     Images  Instances      Box(P          R      mAP50
mAP50-95): 100% ━━━━━━━━━━━━ 15/15 5.7it/s 2.6s0.2s
                   all        239        343   5.71e-05     0.0116   2.94e-05
1.26e-05

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       2/15       1.1G      3.145      4.614      3.677          8        640
━━━━━━━━━━━━ 99/99 5.2it/s 18.9s0.2s
                 Class     Images  Instances      Box(P          R      mAP50
mAP50-95): 100% ━━━━━━━━━━━━ 15/15 5.9it/s 2.6s0.2s
                   all        239        343     0.0393     0.0987     0.0318
0.014

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       3/15      1.11G      2.768      3.907        3.1          8        640
━━━━━━━━━━━━ 99/99 4.8it/s 20.4s0.2ss
                 Class     Images  Instances      Box(P          R      mAP50
mAP50-95): 100% ━━━━━━━━━━━━ 15/15 5.5it/s 2.7s0.2s
                   all        239        343      0.576     0.0824     0.0547
0.0242
```

```
      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       4/15      1.11G      2.366      3.054      2.592         11        640
━━━━━━━━━━━━ 99/99 5.5it/s 18.0s0.2s
                 Class     Images  Instances      Box(P          R      mAP50
mAP50-95): 100% ━━━━━━━━━━━━ 15/15 5.7it/s 2.6s0.2s
                   all        239        343      0.498      0.219      0.222
0.135

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       5/15      1.11G      2.035      2.479      2.331         10        640
━━━━━━━━━━━━ 99/99 5.5it/s 17.9s0.2s
                 Class     Images  Instances      Box(P          R      mAP50
mAP50-95): 100% ━━━━━━━━━━━━ 15/15 5.6it/s 2.7s0.2s
                   all        239        343      0.849      0.327      0.385
0.247
Closing dataloader mosaic

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       6/15      1.11G      1.536      2.185      1.947          3        640
━━━━━━━━━━━━ 99/99 5.3it/s 18.5s0.2s
                 Class     Images  Instances      Box(P          R      mAP50
mAP50-95): 100% ━━━━━━━━━━━━ 15/15 5.2it/s 2.9s0.2s
                   all        239        343      0.694      0.414      0.479
0.305

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       7/15      1.11G      1.374      1.807      1.801          5        640
━━━━━━━━━━━━ 99/99 5.6it/s 17.8s0.2s
                 Class     Images  Instances      Box(P          R      mAP50
mAP50-95): 100% ━━━━━━━━━━━━ 15/15 5.7it/s 2.6s0.2s
                   all        239        343      0.717      0.439      0.456
0.313

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       8/15      1.11G      1.227      1.578      1.684          6        640
━━━━━━━━━━━━ 99/99 5.5it/s 18.0s0.2s
                 Class     Images  Instances      Box(P          R      mAP50
mAP50-95): 100% ━━━━━━━━━━━━ 15/15 5.6it/s 2.7s0.2s
                   all        239        343      0.796      0.487      0.505
0.355

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       9/15      1.13G      1.165      1.408      1.594          4        640
━━━━━━━━━━━━ 99/99 5.5it/s 18.0s0.2s
                 Class     Images  Instances      Box(P          R      mAP50
mAP50-95): 100% ━━━━━━━━━━━━ 15/15 5.7it/s 2.6s0.2s
                   all        239        343      0.905      0.487      0.577
0.419

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      10/15      1.14G      1.135      1.291      1.548          4        640
━━━━━━━━━━━━ 99/99 5.4it/s 18.4s0.2s
                 Class     Images  Instances      Box(P          R      mAP50
mAP50-95): 100% ━━━━━━━━━━━━ 15/15 5.7it/s 2.6s0.2s
                   all        239        343      0.866      0.541      0.562
0.417

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      11/15      1.16G       1.08      1.201      1.514          4        640
━━━━━━━━━━━━ 99/99 5.6it/s 17.8s0.2s
```

```
                 Class     Images  Instances      Box(P          R      mAP50
mAP50-95): 100% ━━━━━━━━━━━━━━━━━━━━ 15/15 5.6it/s 2.7s0.2s
                   all        239        343      0.801      0.594      0.566
0.418

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      12/15      1.17G      1.068      1.157      1.487          4        640
━━━━━━━━━━━━━━━━━━━━ 99/99 5.6it/s 17.7s0.2s
                 Class     Images  Instances      Box(P          R      mAP50
mAP50-95): 100% ━━━━━━━━━━━━━━━━━━━━ 15/15 5.7it/s 2.6s0.2s
                   all        239        343      0.882      0.544      0.597
0.454

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      13/15      1.17G     0.9724      1.055      1.403          3        640
━━━━━━━━━━━━━━━━━━━━ 99/99 5.5it/s 18.0s0.2s
                 Class     Images  Instances      Box(P          R      mAP50
mAP50-95): 100% ━━━━━━━━━━━━━━━━━━━━ 15/15 5.8it/s 2.6s0.2s
                   all        239        343      0.811      0.565      0.541
0.402

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      14/15      1.17G     0.9845      1.051      1.407          3        640
━━━━━━━━━━━━━━━━━━━━ 99/99 5.6it/s 17.6s0.2s
                 Class     Images  Instances      Box(P          R      mAP50
mAP50-95): 100% ━━━━━━━━━━━━━━━━━━━━ 15/15 5.8it/s 2.6s0.2s
                   all        239        343      0.819       0.64      0.676
0.512

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      15/15      1.17G     0.9706       1.03      1.377          3        640
━━━━━━━━━━━━━━━━━━━━ 99/99 5.5it/s 18.0s0.2s
                 Class     Images  Instances      Box(P          R      mAP50
mAP50-95): 100% ━━━━━━━━━━━━━━━━━━━━ 15/15 5.7it/s 2.6s0.2s
                   all        239        343      0.777      0.666      0.639
0.478

15 epochs completed in 0.098 hours.
Optimizer stripped from C:
\School\yolo5\runs\detect\roadsigns_scratch_5ep_bs82\weights\last.pt, 6.2MB
Optimizer stripped from C:
\School\yolo5\runs\detect\roadsigns_scratch_5ep_bs82\weights\best.pt, 6.2MB

Validating C:
\School\yolo5\runs\detect\roadsigns_scratch_5ep_bs82\weights\best.pt...
Ultralytics 8.4.24  Python-3.10.20 torch-2.7.1+cu118 CUDA:0 (NVIDIA GeForce R
4060 Laptop GPU, 8188MiB)
YOLOv8n summary (fused): 73 layers, 3,006,428 parameters, 0 gradients, 8.1 GF
                 Class     Images  Instances      Box(P          R      mAP50
mAP50-95): 100% ━━━━━━━━━━━━━━━━━━━━ 15/15 4.8it/s 3.1s0.2s
                   all        239        343      0.818       0.64      0.676
0.512
          trafficlight         25         37      0.902      0.135      0.215
0.0739
                  stop         27         27      0.584      0.778      0.779
0.673
            speedlimit        182        212      0.911       0.92      0.957
0.781
             crosswalk         53         67      0.875      0.729      0.752
0.519
```

```
Speed: 0.3ms preprocess, 2.9ms inference, 0.0ms loss, 2.5ms postprocess per i
Results saved to C:\School\yolo5\runs\detect\roadsigns_scratch_5ep_bs82
```

In [9]:
```
#test 15
best_model = YOLO("runs/detect/roadsigns_scratch_15ep_bs8/weights/best.pt

metrics = best_model.val(
    data="roadsigns.yaml",
    split="test",
    imgsz=640,
    device=0,
    name="roadsigns_scratch_15ep_bs8_test"
)

print("P        =", metrics.box.mp)
print("R        =", metrics.box.mr)
print("mAP50    =", metrics.box.map50)
print("mAP50-95 =", metrics.box.map)
```

```
Ultralytics 8.4.24  Python-3.10.20 torch-2.7.1+cu118 CUDA:0 (NVIDIA GeForce R
4060 Laptop GPU, 8188MiB)
YOLOv8n summary (fused): 73 layers, 3,006,428 parameters, 0 gradients, 8.1 GF
val: Fast image access  (ping: 0.20.1 ms, read: 212.2110.1 MB/s, size: 260.6
val: Scanning C:\School\yolo5\road-sign-detection\labels\test.cache... 243 im
0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 243/243  0.0s
                 Class     Images  Instances      Box(P          R      mAP50
mAP50-95): 100% ━━━━━━━━━━━━ 16/16 3.4it/s 4.8s0.2s
                   all        243        343      0.746      0.599
0.663        0.5
          trafficlight         32         47      0.289     0.0213      0.133
0.0549
                  stop         29         29      0.854      0.724      0.791
0.627
            speedlimit        188        218      0.917      0.915      0.947
0.775
             crosswalk         45         49      0.923      0.735      0.779
0.542
Speed: 3.5ms preprocess, 4.3ms inference, 0.0ms loss, 2.6ms postprocess per i
Results saved to C:\School\yolo5\runs\detect\roadsigns_scratch_15ep_bs8_test
P        = 0.7458187448537563
R        = 0.5986241284962524
mAP50    = 0.6626047212716392
mAP50-95 = 0.49973497641621967
```

In [10]:
```
#Q4 you can run this code if you have pandas
"""
import pandas as pd
import matplotlib.pyplot as plt

# 1) Put your lab results into a table
df = pd.DataFrame([
    {
        "Setting": "Pretrained, no training",
        "Init": "yolov8n.pt",
        "Epochs": 0,
        "P": 0.0004,
        "R": 0.0266,
        "mAP50": 0.000226,
        "mAP50-95": 0.0000826,
    },
```

```
    {
        "Setting": "Pretrained + fine-tuning",
        "Init": "yolov8n.pt",
        "Epochs": 5,
        "P": 0.938,
        "R": 0.889,
        "mAP50": 0.928,
        "mAP50-95": 0.773,
    },
    {
        "Setting": "Scratch",
        "Init": "yolov8n.yaml",
        "Epochs": 5,
        "P": 0.695,
        "R": 0.327,
        "mAP50": 0.363,
        "mAP50-95": 0.217,
    },
    {
        "Setting": "Scratch",
        "Init": "yolov8n.yaml",
        "Epochs": 15,
        "P": 0.746,
        "R": 0.599,
        "mAP50": 0.663,
        "mAP50-95": 0.500,
    },
])

# 2) Print it in the console
print(df.to_string(index=False))

# 3) Make a picture of the table
fig, ax = plt.subplots(figsize=(12, 2.8))
ax.axis("off")

table = ax.table(
    cellText=df.values,
    colLabels=df.columns,
    loc="center",
    cellLoc="center",
)

table.auto_set_font_size(False)
table.set_fontsize(10)
table.scale(1, 1.6)

plt.tight_layout()

# 4) Save the table picture
plt.savefig("performance_table.png", dpi=300, bbox_inches="tight")
plt.show()"""
```

Out[10]:

```
'\nimport pandas as pd\nimport matplotlib.pyplot as plt\n\n# 1) Put your lab
results into a table\ndf = pd.DataFrame([\n    {\n        "Setting": "Pretra
no training",\n        "Init": "yolov8n.pt",\n        "Epochs": 0,\n
0.0004,\n        "R": 0.0266,\n        "mAP50": 0.000226,\n        "mAP50-95
0.0000826,\n    },\n    {\n        "Setting": "Pretrained + fine-tuning",\n
"Init": "yolov8n.pt",\n        "Epochs": 5,\n        "P": 0.938,\n        "R
0.889,\n        "mAP50": 0.928,\n        "mAP50-95": 0.773,\n    },\n    {\n
"Setting": "Scratch",\n        "Init": "yolov8n.yaml",\n        "Epochs":
5,\n        "P": 0.695,\n        "R": 0.327,\n        "mAP50": 0.363,\n
"mAP50-95": 0.217,\n    },\n    {\n        "Setting": "Scratch",\n        "I
"yolov8n.yaml",\n        "Epochs": 15,\n        "P": 0.746,\n        "R":
0.599,\n        "mAP50": 0.663,\n        "mAP50-95": 0.500,\n    },\n])\n\n#
Print it in the console\nprint(df.to_string(index=False))\n\n# 3) Make a pic
the table\nfig, ax = plt.subplots(figsize=(12, 2.8))\nax.axis("off")\n\ntabl
ax.table(\n    cellText=df.values,\n    colLabels=df.columns,\n    loc="cent
\n    cellLoc="center",\n)\n\ntable.auto_set_font_size(False)
\ntable.set_fontsize(10)\ntable.scale(1, 1.6)\n\nplt.tight_layout()\n\n# 4)
the table picture\nplt.savefig("performance_table.png", dpi=300,
bbox_inches="tight")\nplt.show()'
```

In [ ]:

```python
from pathlib import Path
from ultralytics import YOLO
import matplotlib.pyplot as plt

# Pick a few test images
test_dir = Path("road-sign-detection/images/test")

# Option A: first 4 images
sample_images = sorted(test_dir.glob("*.png"))[:4]

# If your test images are jpg instead, use:
# sample_images = sorted(test_dir.glob("*.jpg"))[:4]

print("Sample images:")
for p in sample_images:
    print(p)

# Load the 4 models/settings
models = {
    "PTNaiveQ1 (0 ep)": YOLO("yolov8n.pt"),
    "Pretrained (5 ep)": YOLO("runs/detect/roadsigns_pretrained_5ep_bs8/w
    "Scratch5 (5 ep)": YOLO("runs/detect/roadsigns_scratch_5ep_bs8/weight
    "Scratch15 (15 ep)": YOLO("runs/detect/roadsigns_scratch_15ep_bs8/wei
}

# Show side-by-side qualitative comparison
n_rows = len(sample_images)
n_cols = len(models)

fig, axes = plt.subplots(n_rows, n_cols, figsize=(4*n_cols, 4*n_rows))

# If only one row, make axes 2D-like
if n_rows == 1:
    axes = [axes]

for i, img_path in enumerate(sample_images):
    for j, (title, model) in enumerate(models.items()):
        results = model(str(img_path))
        annotated_bgr = results[0].plot()   # returns annotated image
        annotated_rgb = annotated_bgr[..., ::-1]  # BGR -> RGB for matplo
```

```
        ax = axes[i][j]
        ax.imshow(annotated_rgb)
        ax.set_title(f"{title}\n{img_path.name}", fontsize=10)
        ax.axis("off")

plt.tight_layout()
plt.savefig("qualitative_results_grid.png", dpi=300, bbox_inches="tight")
plt.show()
```

```
Sample images:
road-sign-detection\images\test\road0.png
road-sign-detection\images\test\road100.png
road-sign-detection\images\test\road103.png
road-sign-detection\images\test\road104.png

image 1/1 C:\School\yolo5\road-sign-detection\images\test\road0.png: 640x448
traffic light, 24.2ms
Speed: 5.0ms preprocess, 24.2ms inference, 5.0ms postprocess per image at sha
3, 640, 448)

image 1/1 C:\School\yolo5\road-sign-detection\images\test\road0.png: 640x448
trafficlight, 20.1ms
Speed: 4.8ms preprocess, 20.1ms inference, 2.5ms postprocess per image at sha
3, 640, 448)

image 1/1 C:\School\yolo5\road-sign-detection\images\test\road0.png: 640x448
speedlimit, 20.7ms
Speed: 4.5ms preprocess, 20.7ms inference, 3.4ms postprocess per image at sha
3, 640, 448)

image 1/1 C:\School\yolo5\road-sign-detection\images\test\road0.png: 640x448
stop, 1 speedlimit, 19.6ms
Speed: 5.3ms preprocess, 19.6ms inference, 2.9ms postprocess per image at sha
3, 640, 448)

image 1/1 C:\School\yolo5\road-sign-detection\images\test\road100.png: 640x64
cake, 27.9ms
Speed: 8.0ms preprocess, 27.9ms inference, 4.7ms postprocess per image at sha
3, 640, 640)

image 1/1 C:\School\yolo5\road-sign-detection\images\test\road100.png: 640x64
speedlimits, 25.0ms
Speed: 9.2ms preprocess, 25.0ms inference, 4.2ms postprocess per image at sha
3, 640, 640)

image 1/1 C:\School\yolo5\road-sign-detection\images\test\road100.png: 640x64
detections), 25.7ms
Speed: 8.5ms preprocess, 25.7ms inference, 2.0ms postprocess per image at sha
3, 640, 640)

image 1/1 C:\School\yolo5\road-sign-detection\images\test\road100.png: 640x64
stops, 25.3ms
Speed: 6.7ms preprocess, 25.3ms inference, 3.7ms postprocess per image at sha
3, 640, 640)

image 1/1 C:\School\yolo5\road-sign-detection\images\test\road103.png: 512x64
stop sign, 25.5ms
Speed: 5.0ms preprocess, 25.5ms inference, 3.4ms postprocess per image at sha
3, 512, 640)

image 1/1 C:\School\yolo5\road-sign-detection\images\test\road103.png: 512x64
speedlimit, 29.7ms
Speed: 6.8ms preprocess, 29.7ms inference, 4.2ms postprocess per image at sha
3, 512, 640)

image 1/1 C:\School\yolo5\road-sign-detection\images\test\road103.png: 512x64
detections), 20.7ms
Speed: 6.8ms preprocess, 20.7ms inference, 1.4ms postprocess per image at sha
3, 512, 640)
```

```
image 1/1 C:\School\yolo5\road-sign-detection\images\test\road103.png: 512x64
stops, 20.3ms
Speed: 4.5ms preprocess, 20.3ms inference, 3.0ms postprocess per image at sha
3, 512, 640)
```

In [ ]: